$$A = \begin{pmatrix} -2 & 1 \\ 0 & a \end{pmatrix}$$

$$\frac{d}{dt}\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 0 & a \end{pmatrix}\begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$$

$\lambda_1 = -2$ $\quad x_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

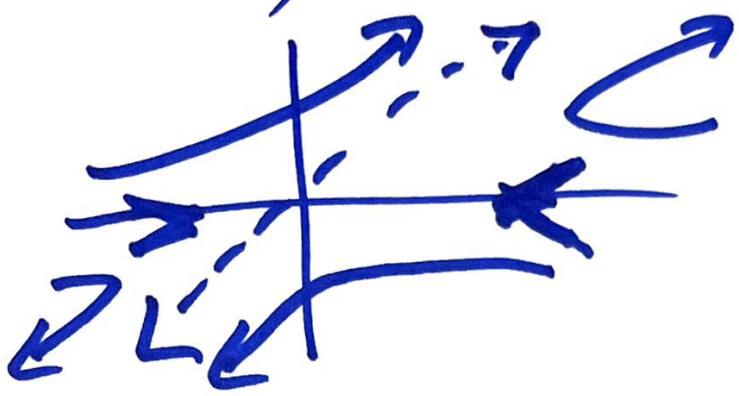
$\lambda_2 = a$ $\quad x_2 = \begin{pmatrix} 1 \\ a+2 \end{pmatrix}$

If $a = -2$, $\lambda_1 = \lambda_2$; $\underline{x}_1 = \underline{x}_2$

$$x(t) = C_1\begin{pmatrix} 1 \\ 0 \end{pmatrix}e^{-2t} + C_2\begin{pmatrix} 1 \\ a+2 \end{pmatrix}e^{at};$$

$\underline{a \neq -2}$

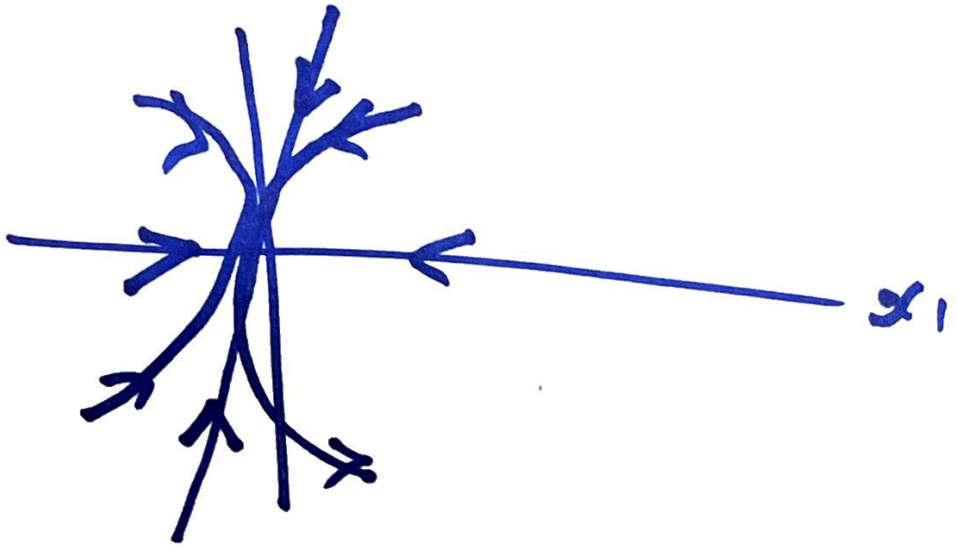
-2- $a > \emptyset$



$x_1 = \begin{pmatrix} 1 \\ c \end{pmatrix}$

$x_2 = \begin{pmatrix} 1 \\ a+2 \end{pmatrix}$

$\emptyset > a > -2$



-3- $a_1 = -2$

$$A = \begin{pmatrix} -2 & 1 \\ 0 & -2 \end{pmatrix} \qquad \lambda_1 = \lambda_2 = -2$$

$$\underline{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\underline{x}(t) = e^{-2t}\left(t \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \underline{B}\right)$$

$$(A - \lambda \underline{I})\underline{B} = \underline{x} \; ; \; x(t) = (\underline{x}t + \underline{B})e^{\lambda t}$$

$\lambda = -2$

$$A - \lambda I = A - (-2)\underline{I} =$$

$$= A + 2I = \begin{pmatrix} -2 & 1 \\ 0 & -2 \end{pmatrix} + 2\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} B_1 \\ B_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$\underline{B} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ $\qquad B_2 = 1, \; B_1 = 0$

$$x(t) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-2t}$$

$$x(t) = e^{-2t}\begin{pmatrix} t \\ 1 \end{pmatrix}$$

-4

$$x(t) = C_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-2t} + C_2 \begin{pmatrix} t \\ 1 \end{pmatrix} e^{-2t}$$

$$x(t) = \begin{pmatrix} t \\ 1 \end{pmatrix} e^{-2t}$$

$$\frac{d}{dt} x(t) = e^{-2t} \begin{pmatrix} 1 - 2t \\ -2 \end{pmatrix}$$

$$A \cdot x = \begin{pmatrix} -2 & 1 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} t \\ 1 \end{pmatrix} e^{-2t} = \Bigg] \Theta$$

$$= \begin{pmatrix} -2t + 1 \\ -2 \end{pmatrix} e^{-2t}$$

5. • $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

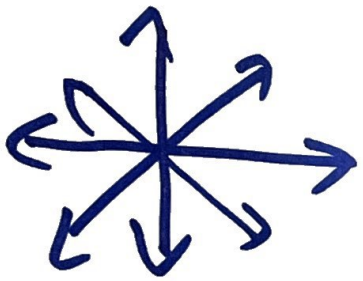
$\lambda_1 = 1; \quad x_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$\lambda_2 = 1 \; ; \; x_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\underline{x}(t) = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^t = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$

$\underline{x}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$

$y(t) = \frac{c_2}{c_1} x(t)$

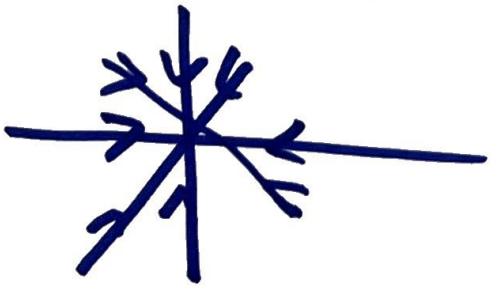


unstable star

• $A = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}$

$\lambda_1 = \lambda_2 = -2$

$x_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad x_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$



6)

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\frac{d}{dt}\,\underline{x}(t) = \frac{d}{dt}\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = 0$$

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

7

$$\frac{d}{dt}\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}\begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$$

$$\tau = a_{11} + a_{22}$$

$$\Delta = a_{11}a_{22} - a_{12}a_{21}$$

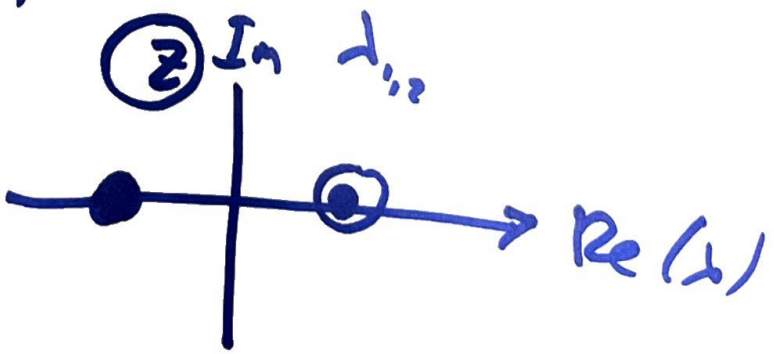
$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$$

- $\Delta < 0$

$\lambda_1 > 0, \quad \lambda_2 < 0$

$\lambda_2 < 0 < \lambda_1$

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}$$



8 $\lambda_{1,2} = \dfrac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$

- $\Delta = 0$: $\lambda_{1,2} = \dfrac{\tau \pm |\tau|}{2} = \{0, \tau\}$

$\sqrt{\tau^2} = |\tau|$



$\tau > 0$ – line of unstable fixed points



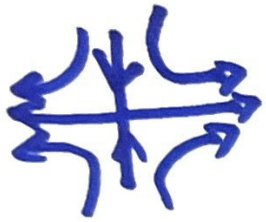
$\tau < 0$ line stable fixed points

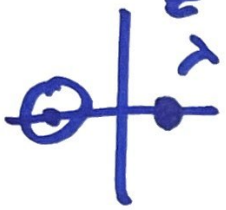
- $\Delta = \tau = 0$, $\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$

plane of fixed points

8 $\Delta < \frac{\tau^2}{4}$

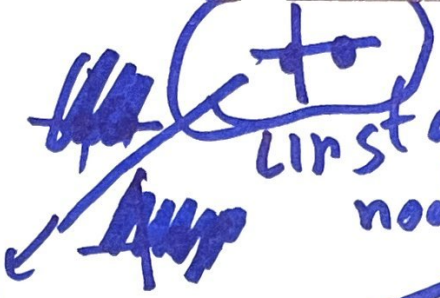
Saddle point



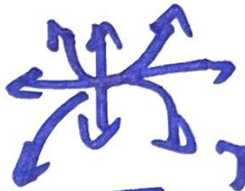


τ

line of unstable FP

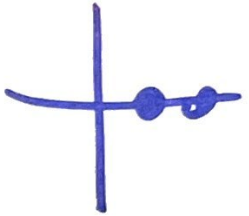


unstable node



$\tau^2/4$

unstable star, deflective node



unstable spiral



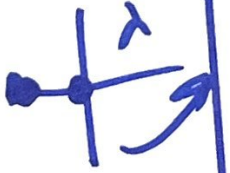
center

Δ

plane of fixed points

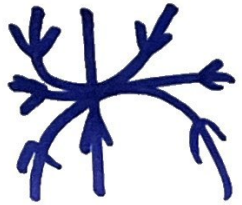
line

stable FP



stable star, deflective node $\tau^2/4$

Stable node



9 $\lambda_{1,2} = \frac{T \pm \sqrt{T^2 - 4\Delta}}{2}$

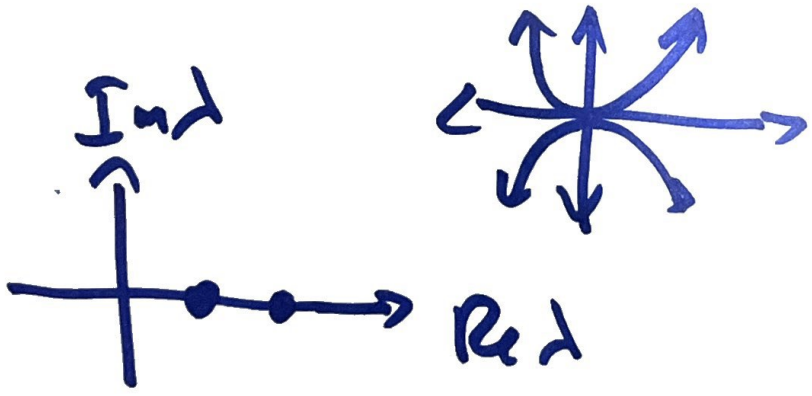
- $\Delta > 0$, $T > 0$, $T^2 > 4\Delta$; $\Delta = \frac{T^2}{4}$ curve on diagram distinguishing scenario's

$T^2 - 4\Delta > 0$

$T > \sqrt{T^2 - 4\Delta}$

$\lambda_2 > \lambda_1 > 0$

unstable node



5 - $T^2 = 4\Delta$ $\lambda_1 = \lambda_2 = T/2$

unstable star, deffective node

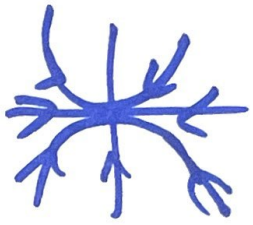
10

- $\tau < 0$, $\tau^2 > 4\Delta$

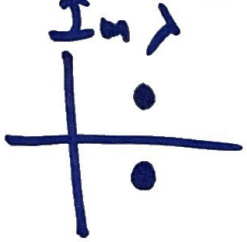
$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$$

$$\lambda_1 < \lambda_2 < 0$$

Stable node



- $\tau > 0$, $\Delta > 0$; $4\Delta > \tau^2$



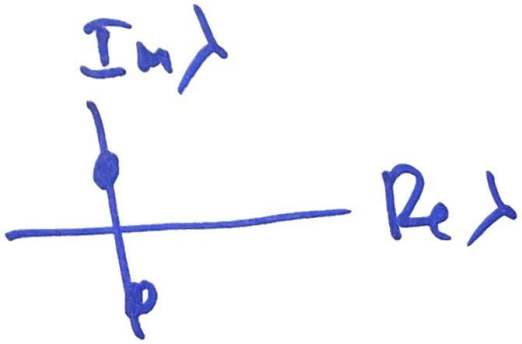


Unstable Spiral

- $\tau = 0$

~~$\lambda_1 = \lambda_2$~~

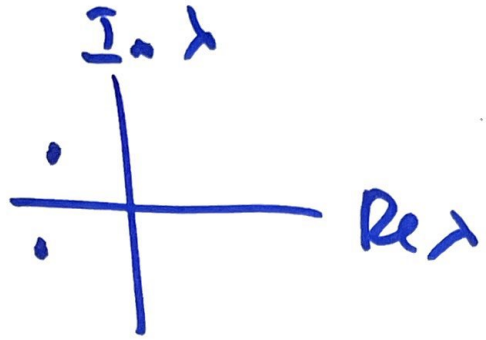
$$\lambda_{1,2} = \pm i\sqrt{|\Delta|} \cdot 2$$



$$\begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix} \quad T = 0 \quad \Delta = 0$$

12

$T < 0, \quad \Delta > 0 \quad 4\Delta > T^2$



Stable spiral



13. $R(t)$ - Romeo

$J(t)$ - Juliet

$$\frac{d}{dt} R(t) = a_{11} R(t) + a_{12} J(t)$$

$$\frac{d}{dt} J(t) = a_{21} R(t) + a_{22} J(t)$$

a_{11} - Romeo reacting to his feelings

a_{22} J her

a_{12} Reaction of Romeo to Juliet

a_{21} Juliet to

$$\frac{d}{dt}\begin{pmatrix} R(t) \\ J(t) \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}\begin{pmatrix} R(t) \\ J(t) \end{pmatrix}$$

Both R, J are out of touch with their feelings

$a_{11} = a_{22} = \emptyset$.

$a_{12} = 1$; He loves J when J loves him
hates hates

$R(t) > \emptyset$ love
$R(t) = \emptyset$ indifferent
$R(t) < \emptyset$ hates

$a_{21} = -1$

$$\frac{d}{dt}\begin{pmatrix} R \\ J \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} R \\ J \end{pmatrix}$$

$T = \emptyset$; $\Delta = 1$ - center



$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

(1) R loves J
J loves R

(2) R loves J
J hates R

(3) R hates J
J hates R

(4) R hates J
J loves R